Q2:

Public → 203.0.113.13
Private → 192.168.10.0/24

(a)

	Public IP/Port	Private IP/Port
1-	203.0.113.13/P1	192.168.10.25/51500
2-	203.0.113.13/P2	192.168.10.30/62000
	$P1 \neq P2$	

(b)

NAT looks up the public port and the destination of incoming packet, replaces it wit private IP as mapped in the NAT table correspondly to the destination and forward to the internal network.

(c)

No, the SSH connection will fail. Because NAT blocks unknown inbound connections unless port forwarding is configured or static NAT (1:1) is configured. There is not existing NAT mapping for public port. Hence, since no port forwarding is configured for SSH, the NAT router cannot map the inbound traffic to the internal server.

(d)

I.

- For events 1 & 2, the controller ~~ad~~ adds outbound flow entries that match the internal source IP/port and destination IP/port, then rewrite the source IP & port to the public NAT IP & assigned port & forward the packet out the WAN.
- For event 3, the controller ~~add~~ installs a reverse flow entry that matches the public IP and NAT port in the incoming packet, the rewrites the destination IP & port bck to the orignal host & forwards packets to LAN.

~~II.~~

~~Event~~

II: Event → E

• **Match**

	Source IP	Source Port	Protocol	Destination IP	Destination Port
E(1)	192.168.10.25	51500	TCP	172.217.160.110	80
E(2)	192.168.10.30	62000	UDP	52.45.33.120	5060
E(3)	203.0.113.15	60001	TCP	—	—

• **Actions:**

Event	Rewrite IP	Port	Forward to
E(1)	Source IP: 203.0.113.15	Source: 60001	internet
E(2)	Source IP: 203.0.113.15	Source: 60002	internet
E(3)	Destination IP: 192.168.10.25	Destination: 51500	internal network

III:

Generalized forwarding allows matching on multiple fields and performing actions like rewrite and forward, enabling complex policies like NAT. This is more flexible than traditional destination-based forwarding, which only looks at destination IP.